

Ou's Proof of the BDPP Theorem for Compact Kähler Manifolds

Kähler MMP

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This note gives a complete proof of Ou's BDPP theorem.

Theorem 0.1 ([Ou25, CP25]). *A compact Kähler manifold is uniruled if and only if K_X is not pseudo-effective.*

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1 Overview of the proof

Let us first sketch the idea of the proof. We prove the theorem by contradiction.

Step 1. (Construct meromorphic fibration $f: X \dashrightarrow Y$) Assume that K_X is not pseudo-effective and X is not uniruled, then a destabilizing subsheaf of T_X induces a meromorphic fibration

$$X \dashrightarrow Y,$$

such that $\dim X > \dim Y > 0$ (by Theorem 3.2). Note that, since we assume X is not projective, there exists a nonzero $\sigma \in H^0(X, \Omega_X^2)$. By Theorem 3.2, we have $H^0(Y, \Omega_Y^2) \neq 0$ as well.

Step 2. (Apply relative Albanese reduction) We then resolve the indeterminacy of $f: X \dashrightarrow Y$, and apply the relative Albanese reduction repeatedly to $f: X \rightarrow Y$ (cf. Theorem 5.1), so that we get the diagram below

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow p & \nearrow q \\ & W & \end{array}$$

such that the following properties hold:

- (a) Let F be a general fiber of p . Then $\dim F > 0$. In addition, either $H^1(F, \mathcal{O}_F) = \{0\}$ (Albanese dimension 0) or F has maximal Albanese dimension.
- (b) A general fiber of $q: W \rightarrow Y$ is not uniruled.

Step 3. (Apply pseudo-effectiveness in family) We then apply the pseudo-effectiveness of adjoint classes in families (Theorem 4.1) to show that Y is uniruled. More precisely, since we assume that X is not uniruled, the general fiber F is also not uniruled. By induction on the dimension, this implies that K_F is pseudo-effective. Therefore, by Theorem 4.1, the relative canonical divisor $K_{X/W}$ is pseudo-effective. Consequently, the only remaining possibility is that K_W is not pseudo-effective. Again by induction on the dimension, W is uniruled, and hence Y is uniruled.

Step 4. (Prove rational connectivity of Y , deduce contradiction) We claim, furthermore, that Y is rationally connected. For if Y is not rationally connected, we apply the MRC fibration to Y and get

$$Y \dashrightarrow V$$

and apply the same argument to

$$X \dashrightarrow V,$$

we deduce that V is uniruled, contradicting [GHS03]. So Y is rationally connected.

In particular, $H^0(Y, \Omega_Y^2) = 0$, which contradicts $H^0(Y, \Omega_Y^2) \neq 0$ established above. Hence K_X must be pseudo-effective.

Note that the technical core lies in Step 1 (cf. Theorem 3.2), which reduces to constructing a quasi-psh function with large restricted Lelong number.

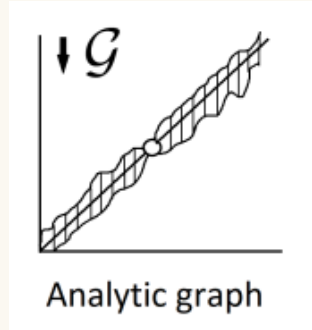
2 Step 1. Produce quasi-psh function with large restricted Lelong number

2.1 Setting

Let X be a compact Kähler manifold, and $C \subset \hat{X}$ be a closed irreducible submanifold. Let S_0 be an irreducible, locally closed submanifold containing a Zariski open subset C° of C .

Let us draw a picture to illustrate the setting: In the picture below, let Y be a compact Kähler manifold, and $\mathcal{F}$ be a foliation on Y . Let X be the product space $X = Y \times Y$ and C is the diagonal. We then keep only the regular part Y_o of the foliation $\mathcal{F}$. Let $X_o = Y_o \times Y_o$ and set $C_o = C \cap X_o$.

We then define a foliation $\mathcal{G} = p_2^{-1}(\mathcal{F}) \cap p_1^{-1}(0)$ on X , then the local leaf of $\mathcal{G}$ along C_o defines a locally closed submanifold S_o as the shaded region below shows.



Our goal is to give an algebraicity criterion for the foliation that guarantees $\dim S^o = \dim \overline{S^o}^{\text{Zar}}$.

2.2 Ou's approach to constructing a quasi-psh function with large restricted Lelong number

Theorem 2.1. *Let (X, ω) be a compact Kähler manifold, let $C \subseteq X$ be a closed irreducible submanifold, and let $S_0 \subseteq X$ be an irreducible locally closed subvariety containing a Zariski open subset $C_0 \neq \emptyset$ of C . We make the following assumptions.*

- (i) S_0 is Zariski dense in X ,
- (ii) $\dim S_0 = \dim C + 1$,
- (iii) The prime divisors in C contained in $C \setminus C_0$ form an exceptional family,
- (iv) There is a Zariski open subset C_1 of C_0 whose complement has codimension at least 2, such that S_0 is smooth around C_1 . Furthermore, S_0 extends formally along C .

Then for any $\lambda > 0$, there is a ω -psh function φ satisfying the following properties.

- (1) φ has analytic singularities. In addition, φ can be locally written as

$$\varphi = \frac{1}{2m} \log \left(|g_1|^2 + \cdots + |g_r|^2 \right) + O(1)$$

where $m \in \mathbb{Z}_{>0}$, and $g_1, \dots, g_r$ are local generators of a coherent ideal sheaf $\mathcal{I}$ on X .
 (2) For all points $y \in C_1$, we have $v(\varphi|_{S_0}, y) > \lambda$.

💡 Idea of proof. We divide the proof of this main theorem into three steps.

Step 1. (Produce quasi-psh function with large restricted Lelong number). We use Demailly's mass concentration technique combined with Demailly's regularization technique to show that there exists a quasi-psh function (with analytic singularities) ψ_3 with large restricted Lelong number

$$v(\psi_3|_{S_0}, x) \geq \frac{1}{2} \cdot \frac{1}{m_2} \cdot (\mu m_2 - m_1 + 1 - n) = \frac{1}{2}\mu - \frac{m_1 + n - 1}{2m_2},$$

where ψ_3 is of the form

$$\psi_3 = \frac{1}{4m_2} \log(|h_1|^2 + \dots + |h_r|^2) + O(1).$$

Step 2. (Take a blowup and produce a closed positive current upstairs) We then take a blowup $\rho: \widehat{X} \rightarrow X$, and consider the closed positive current

$$T' := \rho^* \left(\omega + \frac{\sqrt{-1}}{\pi} \partial \bar{\partial} \psi_3 \right) - \frac{\zeta}{2m_2} [E],$$

where E is the unique component of the ρ -exceptional divisor that meets the strict transform of S and ζ is the minimal vanishing order of $h_1 \circ \rho, \dots, h_r \circ \rho$ along E . The current is positive by Lemma 7.2.

Step 3. (Pushforward the current onto X , deduce contradiction by boundedness of Lelong number) We then restrict the current to C and push it forward

$$T_C = \rho_*(T'|_C).$$

It can be shown that

$$\{T_C\} = \{\omega|_C\} - \frac{1}{2m_2} \zeta \cdot c_1(\mathcal{N}) + \delta$$

and

$$v(T_C, x) = v(T'|_{C'}, x') \geq v(T'|_{S'_0}, x') = v(\psi_3|_{S_0}, x) - \frac{1}{2m_2} \zeta \geq \frac{\mu}{3} - K$$

which contradicts the boundedness result on Lelong numbers (Lemma 7.8).

2.3 Cao–Păun’s approach to constructing quasi-psh function with large restricted Lelong number

One of the central lemmas we use is the following, which allows us to produce a quasi-psh function with large restricted Lelong number. We mainly follow the purely analytic proof provided by [CP25].

Theorem 2.2 ([CP25, Proposition 4.4]). *Let X be a compact Kähler manifold, and let $C \subset X$ be a compact submanifold. Let S_0 be an irreducible, locally closed submanifold containing C_0 , where C_0 is a dense open subset of C satisfying $\text{codim}_C(C \setminus C_0) \geq 2$. Let M be the Zariski closure of S_0 in X .*

Let $\pi : \widehat{X} \rightarrow X$ be the blow-up of C in X . Let $\widehat{S}_0$ be the strict transform of S_0 , and let D be the exceptional divisor of π . Define $E_0 := \widehat{S}_0 \cap D$. Let $\omega_{\widehat{X}}$ be a Kähler metric on $\widehat{X}$.

If $\dim M > \dim S_0$, then for any $m \in \mathbb{N}$, there exists a $\omega_{\widehat{X}}$ -psh function φ_m on $\widehat{X}$ such that $\varphi_m|_{\widehat{S}_0}$ has analytic singularities and $v(\varphi_m|_{\widehat{S}_0}, E_0) \geq m$.

Idea of proof. We divide the proof into three steps:

Step 1. (Construction of quasi-psh function on the smooth model of $\widehat{M} = \pi_*^{-1}(M)$). To do this, we take a modification $\widetilde{M} \rightarrow \widehat{M}$ and construct a quasi-psh function with large restricted Lelong number along $\widetilde{S}_0$ (more details of the proof appear in Proposition 2.3). The diagram is shown below:

$$\begin{array}{ccccc}
 \widetilde{y}_0 \in \widetilde{E}_0 & \hookrightarrow & \widetilde{S}_0 & \hookrightarrow & \widetilde{X} \\
 \downarrow p & & \downarrow p & & \downarrow p \\
 \widehat{y}_0 \in E_0 & \hookrightarrow & \widehat{S}_0 & \hookrightarrow & \widehat{X} \\
 \downarrow \pi & & \downarrow \pi & & \downarrow \pi \\
 y_0 \in C_0 & \hookrightarrow & S_0 & \hookrightarrow & X
 \end{array}$$

Step 2. (Using the extension theorem to extend the quasi-psh function onto the whole birational model of $\widehat{X}$). We then use Collins–Tosatti’s extension theorem to extend the quasi-psh function onto the whole $\widehat{X}$. Since the extension does not affect the restriction to $\widetilde{M}$, the restricted Lelong number condition is unchanged.

Step 3. (Push down the associated Kähler current to $\widehat{X}$, bounding the Lelong number). Let

$$\widetilde{T}_m = \omega_{\widetilde{X}} + i\partial\bar{\partial}\widetilde{\varphi}_m$$

be the associated Kähler current. We then push it down to

$$\widehat{T}_m = p_*(\widetilde{T}_m), \quad T_m = \pi_*p_*(\widetilde{T}_m).$$

First we can bound

$$\left| \nu \left(\widehat{T}_m|_{\widehat{S}_0}, \widehat{y}_0 \right) - \nu \left(\widetilde{T}_m|_{\widetilde{S}_0}, \widetilde{y}_0 \right) \right| \leq C_1$$

using Lemma ?? . And therefore $\nu \left(\widehat{T}_m|_{\widehat{S}_0}, \widehat{y}_0 \right) \geq m - C_1$. We then divide the problem into two cases:

- (a) If $\nu \left(\widehat{T}_m|_{\widehat{S}_0}, E_0 \right) \geq \frac{m}{2}$ for infinitely many $m \in \mathbb{N}^*$, then the proposition is proved.
- (b) If $\nu \left(\widehat{T}_m|_{\widehat{S}_0}, E_0 \right) \leq \frac{m}{2}$ for every sufficiently large m , then

$$\nu \left(T_m|_{S_0}, y_0 \right) \geq \frac{1}{K} \cdot \left(\nu \left(\widehat{T}_m|_{\widehat{S}_0}, \widehat{y}_0 \right) - \nu \left(\widehat{T}_m|_{\widehat{S}_0}, E_0 \right) \right) \geq \frac{1}{K} \cdot \left(\frac{m}{2} - C_1 \right)$$

for some uniform constant K independent of m (by Lemma 7.6). And therefore, by Lemma 7.7,

$$\nu \left(T_m|_{S_0}, C_0 \right) \geq \frac{\nu \left(T_m|_{S_0}, y_0 \right) - K_2}{K_1} \geq \tilde{K}_1 m - \tilde{K}_2.$$

Pulling back via p gives the desired quasi-psh function (by Lemma ??).

Our next goal is to complete the construction of a quasi-psh function on the smooth model $\widetilde{M}$ of $\widehat{M}$ with large restricted Lelong number along $\widetilde{S}$.

Lemma 2.3 ([CP25, Proposition 4.1]). *Let X be a compact Kähler manifold, and let ω_X be a Kähler metric on X . Let S_0 be a locally closed submanifold of X such that*

$$\dim S_0 < \dim X$$

and the Zariski closure of S_0 is X .

Fix a point $x \in S_0$. Then, for any $\lambda > 0$, we can find a ω_X -psh function φ on X such that φ has analytic singularities and satisfies $\nu \left(\varphi|_{S_0}, x \right) \geq \lambda$.

💡 Idea of proof. The idea is straightforward. We first construct a current with analytic singularities as a model and denote it

$$f = \frac{1}{m^{\frac{n-1}{n}}} \log \left(|z_1|^2 + |z_2|^{2m} + \cdots + |z_n|^{2m} \right).$$

We then apply Demailly's mass concentration and regularization theorem to find the candidate quasi-psh function with analytic singularities, which is of the form

$$\varphi = \frac{1}{k} \log \sum |g_i|^2 + O(1).$$

In order to control the restricted Lelong number, it suffices to give an explicit bound on the vanishing order of each $g_i|_{z_1=0}$ at 0. To do this, we use the fact that each $g_i \in \mathcal{I}(kCf)$. Since we know the local generators of the ideal sheaf explicitly, this gives the information on the vanishing order of g_i .

Proof. ■

Lemma 2.4 ([CP25, Lemma 4.5]). *The same setting as in the proof of Theorem 2.2. There exists a constant C_1 independent of m such that*

$$\left| \nu \left(\widehat{T} \Big|_{\widehat{S}_0}, \widehat{y}_0 \right) - \nu \left(\widetilde{T} \Big|_{\widetilde{S}_0}, \widetilde{y}_0 \right) \right| \leq C_1.$$

Proof. ■

2.4 Comparison of Ou's approach and Cao-Păun's approach

As noted in [CP25], that paper provides a purely analytic proof.

3 Step 2. Algebraicity criterion for the foliations

3.1 Ou's algebraicity of foliation when conormal sheaf is not pseudo-effective

3.2 Cao-Păun's Algebraicity of foliation when conormal sheaf $\mathcal{N}_{C_0/S_0}^*$ is not pseudo-effective

In this step, we mainly follow the approach of [CP25].

Theorem 3.1 ([CP25, Theorem 1.3]). *Let X be a compact Kähler manifold, and let $C \subset X$ be a compact submanifold. Let S_0 be an irreducible, locally closed submanifold containing C_0 , where C_0 is a dense open subset of C satisfying $\text{codim}_C(C \setminus C_0) \geq 2$. Let M be the Zariski closure of S_0 in X .*

If the conormal bundle $\mathcal{N}_{C_0/S_0}^$ is not pseudo-effective, then $\dim M = \dim S_0$.*

💡 **Idea of proof.** Assume by contradiction that $\dim M > \dim S_0$. Let us consider the blowup $\widehat{X} \rightarrow X$ along C , with strict transform $\widehat{S}_0$, and D the exceptional divisor such that $E_0 = D \cap \widehat{S}_0$. First by Theorem 2.2, for each $m \geq 1$ there exists a $\omega_{\widehat{X}}$ -psh function

φ_m such that

$$a_m := v\left(\varphi_m|_{\widehat{S}_0}, E_0\right) \geq m.$$

Therefore, there exists a sequence of positive currents

$$T_m := \omega_{\widehat{X}} + i\partial\bar{\partial}\varphi_m \geq 0$$

such that

$$G_m := T_m|_{\widehat{S}_0} - a_m [E_0]$$

are still positive. In particular, this implies

$$c_1(G_m|_{E_0}) = c_1\left(\omega_{\widehat{X}}|_{E_0} + a_m \cdot \mathcal{O}_{\mathbb{P}(\mathcal{N}_{C_0/S_0}^*)}(1)\right) \quad \text{on } E_0.$$

This means that $\mathcal{N}_{C_0/S_0}^*$ is pseudo-effective.

3.3 Ou's algebraicity criterion when minimal slope is positive

We now prove the algebraicity criterion for foliations.

Theorem 3.2 ([Ou25, Theorem 1.4]). *Let X be a compact Kähler manifold of dimension n , and let $\mathcal{F}$ be a foliation on X .*

- (1) *If $\mathcal{F}^*$ is not pseudo-effective, then $\mathcal{F}$ is induced by a meromorphic map.*
- (2) *In particular, if the minimal slope satisfies $\mu_{\alpha, \min}(\mathcal{F}) > 0$ for some movable class $\alpha \in H^{n-1, n-1}(X, \mathbb{R})$, then $\mathcal{F}$ is induced by a meromorphic map.*

💡 Idea of proof. To prove this, we apply the algebraicity criterion to the analytic graph of the foliation. To be more precise, we set $C' := X$ and $X' = X \times X$, and the regular locus of the foliation $\mathcal{F}$ to be X^o and $C'_0 = C' \cap (X^o \times X^o)$. We define

$$\mathcal{G} = p_2^{-1}(\mathcal{F}) \cap p_1^{-1}(0),$$

to be a foliation on X' , so that the local leaf of $\mathcal{G}$ defines the locally closed submanifold S'_0 . Since

$$\mathcal{F}^*|_{X^o} \cong \mathcal{N}_{C'_0/S'_0}^*,$$

if $\mathcal{F}^*$ is not pseudo-effective, then neither is $\mathcal{N}_{C'_0/S'_0}^*$, and hence

$$\dim M = \dim S_0.$$

We finally apply Lemma 8.1, so that $\mathcal{F}$ induces a meromorphic map.

Combining all of the above, we have the following theorem.

Theorem 3.3 ([Ou25]). *Let X be a compact Kähler manifold of dimension n and let $\alpha \in H^{n-1,n-1}(X, \mathbb{R})$ be a movable class. Assume that $\mu_\alpha(T_X) > 0$, and that there is some nonzero closed holomorphic 2-form $\sigma \in H^0(X, \Omega_X^2) \cong H^{2,0}(X, \mathbb{C})$. Let $\mathcal{K}$ be the kernel of the morphism*

$$\lrcorner \sigma : T_X \rightarrow \Omega_X^1$$

defined by the contraction with σ . Let $\mathcal{F} \subseteq \mathcal{K}$ be the maximal destabilizer with respect to α . Then $\mathcal{F}$ is a foliation induced by a meromorphic map $f : X \dashrightarrow Y$.

💡 **Idea of proof.**

4 Step 3. Pseudo-effectiveness of the adjoint class in families

In this step, we present pseudo-effectiveness of the adjoint class in families.

Theorem 4.1 ([Ou25]). *Let $f : X \rightarrow Y$ be a fibration between compact Kähler manifolds. We assume the following conditions.*

- (1) $H^1(F, \mathcal{O}_F) = \{0\}$ for a general fiber F of f .
 - (2) K_F is pseudoeffective.
 - (3) f is smooth over an open subset of Y whose complement is a snc divisor.
 - (4) *There is a Kähler form ω on X and a closed holomorphic 2-form $\tau \in H^0(Y, \Omega_Y^2)$, such that the cohomology class $\{\omega + f^*\tau + f^*\bar{\tau}\}$ belongs to the image of $H^2(X, \mathbb{Z})$.*
- Then $K_{X/Y}$ is pseudo-effective.*

💡 **Idea of proof.**

Theorem 4.2. *Let $f : X \rightarrow Y$ be a fibration between compact Kähler manifolds. Assume that general fibers F of f have maximal Albanese dimension. Then $\omega_{X/Y}$ is pseudoeffective.*

Proof. ■

5 Step 4. Relative Albanese reduction

Once we have the meromorphic fibration from Theorem 3.2, we factorize it into two parts:

- (1) The first part is a fibration with Albanese dimension 0 or it has maximal Albanese

dimension; (2) General fibers of the second part are not uniruled. (More about relative Albanese maps can be found in my note [Note-I.2. Albanese map with applications](#)).

Theorem 5.1 ([Ou25, Proposition 8.6]). *Let $f : X \rightarrow Y$ be a fibration between compact complex analytic varieties, such that $\dim X > \dim Y$ and that X is a Kähler manifold. Then, up to blowing up X , there is a factorization of f into fibrations $p : X \rightarrow W$ and $q : W \rightarrow Y$, such that the following properties hold.*

(1) W is a compact Kähler manifold.

(2) There is a Zariski open subset $W^\circ \subseteq W$, whose complement is a simple normal crossing divisor, such that p is smooth over W° .

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow p & \nearrow q \\ & W & \end{array}$$

(3) **(First part of the fibration has maximal Albanese dimension or Albanese dimension 0)** Let F be a general fiber of p . Then $\dim F > 0$. In addition, either $H^1(F, \mathcal{O}_F) = \{0\}$ (Albanese dimension 0) or F has maximal Albanese dimension. (Note that since the fibration is nontrivial, i.e., $\dim F > 0$, the dimension drops along $p : X \rightarrow W$),

(4) **(General fibers of the second part are not uniruled)** General fibers of q are not uniruled. (In particular if Y is not uniruled then W is not uniruled).

💡 Idea of proof. The idea is to repeatedly apply the relative Albanese reduction until condition (3) is satisfied. We focus on conditions (3) and (4), since (1) and (2) trivially hold after blowing up X and W once (3) and (4) hold.

Assume (3) does not hold. Since X is Kähler and f is a fibration, the relative Albanese reduction exists by [GPR94, Theorem 3.27]:

$$X \xrightarrow{\alpha} \text{Alb}(X/Y) \rightarrow Y.$$

We then take the image of the relative Albanese map $\alpha : X \rightarrow \text{Alb}(X/Y)$ as $Z = \alpha(X)$. Since (3) does not hold, this implies that $\dim Z < \dim X$. Since Z is an abelian variety (or a quotient thereof), the general fiber of $Z \rightarrow Y$ cannot be uniruled. Taking a Stein factorization on

$$X \rightarrow W \rightarrow Z,$$

we can guarantee that the first factor is a contraction morphism.

Since $\dim X > \dim Z$ (dimension strictly drops), after finitely many steps of the reduction, (3) and (4) hold.

6 Step 5. Finish the proof of BDPP conjecture for compact Kähler manifold

Having introduced the preparations in the previous steps, we can now complete the proof of Ou's BDPP theorem for compact Kähler manifolds.

Theorem 6.1 ([Ou25, CP25]). *A compact Kähler manifold is uniruled if and only if K_X is not pseudo-effective.*

💡 Idea of proof. First note that a uniruled variety has non-pseudo-effective K_X . The nontrivial part is the converse direction, i.e., if K_X is not pseudo-effective, then X is uniruled. We prove it by contradiction and assume that X is not uniruled.

Step 1. (Construct meromorphic fibration whose base Y is uniruled). We first show that if K_X is not pseudo-effective then $\mu_\alpha(T_X) > 0$; we then apply Theorem 3.2 to find a meromorphic fibration induced by T_X , and denote it

$$X \dashrightarrow Y.$$

We then apply the relative Albanese Theorem 5.1 and factorize it into

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow p & \nearrow q \\ & W & \end{array}$$

We then apply Theorem 4.1 to show that $K_{X/W}$ is pseudo-effective. Since we assume that K_X is not pseudo-effective, K_W is not pseudo-effective. By induction on dimension (since $\dim W < \dim X$), W is uniruled, and by Theorem 5.1 applied again, Y is uniruled.

Step 2. (Prove Y is rationally connected). Assume Y is not rationally connected. We then apply the MRC fibration to the base $Y \rightarrow V$ (resolving the indeterminacy).

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow \tilde{f} & \downarrow g \\ & & V \end{array}$$

By Gabber–Harris–Starr ([GHS03]), V is not uniruled. By a Leray spectral sequence argument,

$$H^0(Y, \Omega_Y^2) = H^0(V, \Omega_V^2).$$

Hence we can invoke Theorem 5.1 again and run the same argument as in Step 1 to show that V is uniruled. This contradicts $H^0(Y, \Omega_Y^2) \neq 0$.

7 Appendix: Lelong number under birational changes

In this section, we study the Lelong number under birational modifications or blowups.

First is an easy observation.

Lemma 7.1. *When T is a positive current with analytic singularities around a point x in a complex manifold X . If $\rho : X' \rightarrow X$ is the blowup of x , then $R := \rho^*T - v(T, x)[E]$ is a positive current, and a local potential of it is bounded around a general point of E .*

Proof. ■

A similar situation holds when we blow up a curve C and consider the restricted Lelong number $v(\varphi|_S, x)$ along a general point of C .

Lemma 7.2. *Let $C \subseteq S \subseteq X$ be germs of complex manifolds around a point o . Assume that $\dim S = \dim C + 1$.*

Let $\varphi = \frac{1}{2} \log(|g_1|^2 + \cdots + |g_r|^2)$ such that $g_1, \dots, g_r$ are holomorphic functions on X . Assume that there is at least one g_i which is nonzero at general points of S . Suppose that the restricted Lelong number $v(\varphi|_S, x) = l$ for a general point $x \in C$.

Then there is a projective bimeromorphic morphism $\rho : X' \rightarrow X$ satisfying the following properties.

(1) ρ is obtained by firstly blowing up C , then blowing up the intersection of the strict transform of S and the exceptional locus several times.

The strict transform S' of S in X' meets the ρ -exceptional divisor transversally at a unique irreducible component E .

(2) If C' is the intersection of S' and E , and if $o' \in C'$ is the point lying over $o \in C$, then

$$\varphi \circ \rho = l \cdot \log |h| + \frac{1}{2} \log v$$

in a neighborhood of o' , where h is a local generator of $\mathcal{O}_{X'}(-E)$, and v is a finite sum of squares of absolute values of holomorphic functions.

Moreover, v is not identically zero along C' .

Remark 7.3. Using the Poincaré–Lelong formula, we know that

$$\frac{\sqrt{-1}}{\pi} \partial \bar{\partial} \log |h| = [E].$$

So that

$$\frac{\sqrt{-1}}{\pi} \partial \bar{\partial} \rho^*(\varphi) - l[E] = \frac{\sqrt{-1}}{2\pi} \partial \bar{\partial} \log v \geq 0,$$

which defines a closed positive current.

Proof. ■

One can compare the Lelong number under pullback.

Proposition 7.4. *Let $f : X \rightarrow Y$ be a surjective morphism between two compact complex manifolds. Given a closed $(1, 1)$ -current T on the base Y , there exists a constant $C > 0$ such that*

$$Cv(f^*T, x) \leq v(T, y) \leq v(f^*T, x),$$

for all $x \in X$ and $y = f(x)$.

One can prove that the Lelong number does not change under pullback via blowup along a submanifold.

Lemma 7.5. *Let $\pi : \widehat{X} \rightarrow X$ be the blowup along Y . Let $x \in X$. If φ is a quasi-psh function on X with analytic singularities, then*

$$v(\varphi, x) = v(\pi^*\varphi, \pi^{-1}(x)), \quad \text{and} \quad v(\varphi, Y) = v(\pi^*\varphi, \pi^{-1}(Y)).$$

Lemma 7.6 ([CP25, Lemma 2.2]). *Let X be a complex manifold, and let $Y \subset X$ be a submanifold. Let $\pi : \widehat{X} \rightarrow X$ be the blowup of Y , and let $E \subset \widehat{X}$ be the exceptional divisor. Let $\widehat{T} \geq 0$ be a positive $(1, 1)$ -current on $\widehat{X}$ with analytic singularities. Let $\widehat{x} \in E$ and set $x := \pi(\widehat{x})$. Define $T := \pi_*(\widehat{T})$. Then*

$$C \cdot v(T, x) \geq v(\widehat{T}, \widehat{x}) - v(\widehat{T}, E)$$

for some uniform constant C depending only on X and Y .

Proof. ■

Lemma 7.7 ([CP25, Proposition 4.2]). *Let X be a compact Kähler manifold and $C \subset X$ be a compact submanifold. Let S_0 be a locally closed submanifold containing C_0 , where C_0 is an open subset of C with $\text{codim}_{\mathbb{C}}(C \setminus C_0) \geq 2$.*

We fix a point $y_0 \in C_0$. Let ω_X be a Kähler metric on X .

Then there exist constants K_1 and K_2 depending only on $(\omega_X, y_0, C_0, S_0, X)$ such that, for any ω_X -psh function φ on X , if $\varphi|_{S_0}$ has algebraic singularities, then

$$v(\varphi|_{S_0}, y_0) \leq K_1 \cdot v(\varphi|_{S_0}, C_0) + K_2.$$

Proof. ■

We have the following boundedness result.

Lemma 7.8 ([Ou25, Lemma 6.1]). *Let (X, ω) be a compact Kähler manifold, let $Z \subseteq X$ be a Zariski closed subset, and let $x \in X \setminus Z$ be a point. Assume that:*

- (a) the prime divisors contained in Z form an exceptional family;*
- (b) for any bounded subset $D \subseteq H^{1,1}(X, \mathbb{R})$, there is a constant K such that, for any class $\alpha \in D$, for any real divisor class δ with support in Z , and for any positive closed current T in the class $\alpha + \delta$, the Lelong number $\nu(T, x)$ is bounded above by K .*

8 Appendix: A brief introduction to foliation theory

We next summarize some criteria for a foliation to induce a fibration.

Lemma 8.1 ([Ou25, Lemma 7.5]). *Let X be a compact Kähler manifold and let $\mathcal{F}$ be a foliation on X . Let $X_0 \subseteq X$ be the regular locus of $\mathcal{F}$.*

Then $\mathcal{F}$ is induced by a meromorphic map if (and only if) its analytic graph $S_0 \subseteq X_0 \times X_0$ has the same dimension as its Zariski closure in $X \times X$.

When the foliation is weakly regular it induces a holomorphic fibration.

References

- [CP25] Junyan Cao and M. Păun, *Remarks on relative canonical bundles and algebraicity criteria for foliations in kähler context*, 2025.
- [GHS03] Tom Graber, Joe Harris, and Jason Starr, *Families of rationally connected varieties*, J. Amer. Math. Soc. **16** (2003), no. 1, 57–67.
- [GPR94] H. Grauert, Th. Peternell, and R. Remmert (eds.), *Several complex variables. VII*, Encyclopaedia of Mathematical Sciences, vol. 74, Springer-Verlag, Berlin, 1994, Sheaf-theoretical methods in complex analysis, A reprint of *Current problems in mathematics. Fundamental directions. Vol. 74* (Russian), Vseross. Inst. Nauchn. i Tekhn. Inform. (VINITI), Moscow.
- [Ou25] W.-H. Ou, *A characterization of uniruled compact kähler manifolds*, 2025.